Factors affecting Enzyme Action

08/06/2026

Keywords: Denature, pH, temperature, optimum, reaction, extremophile

DO NOW: Match the key words to the definitions.

EXT: What is the optimum temperature for human amylase enzymes?

Catalyst

Substrate

Product

ENZYME

Metabolism

Active site

Synthesis

The substance that is broken down

A biological catalyst that speed up reactions

The region of the enzyme where the reaction takes place

The process that describes making something new

Something produced during the reaction of an enzyme and substrate

Something that speeds up a reaction

Catalyst

The substance an enzyme acts upon

Substrate

A biological catalyst that speed up reactions

Product

The region of the enzyme where the reaction takes place

ENZYME

Describes how the shape of the substrate fits the shape of the enzyme

lock and Key

The process that describes making something new

Active site

Something produced during the reaction of an Enzyme and Substrate

Synthesis

Something that speeds up a reaction

Catalyst

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Product

ENZYME

lock and Key

Active site

Synthesis

The substance an enzyme acts upon

A biological catalyst that speed up reactions

The region of the enzyme where the reaction takes place

Describes how the shape of the substrate fits the shape of the enzyme

The process that describes making something new

Something produced during the reaction of an Enzyme and Substrate

Something that speeds up a reaction

LOb: Understand the factors that affect the action of an enzyme.

Keywords: Denature, pH, temperature, optimum, reaction, extremophile

Success Criteria:

- Describe what denaturing is and how an enzyme is denatured by temperature
- Explain how pH affects enzyme action

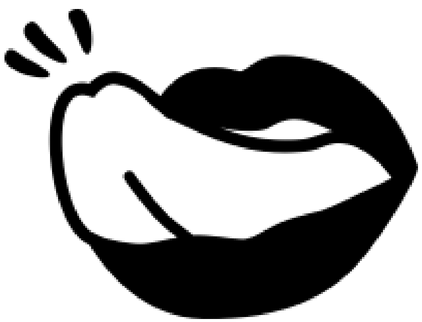
Did you know… **Rubisco** is the most important enzyme as it allows plants to turn carbon dioxide in the air into carbon which allows plants to grow. Without this enzyme life as we know it would not exist. **Rubisco** is also the most abundant enzyme in the world.

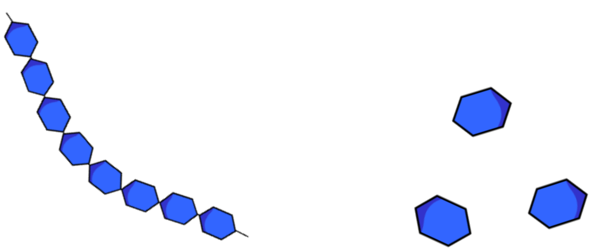
Digestive enzymes are found in these three organs

Amylase

carbohydrase

(Breaks down Carbohydrate)



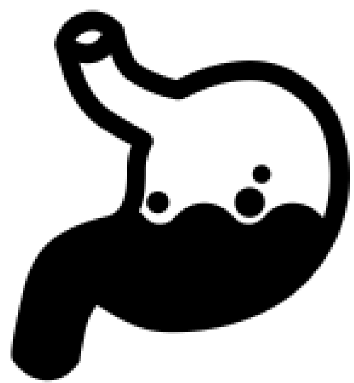


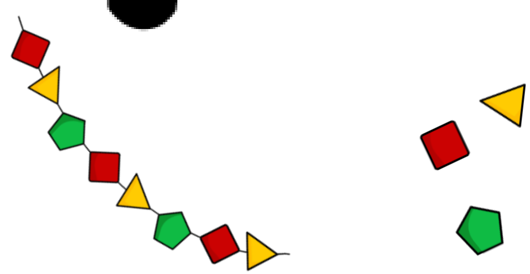
Starch → Glucose

Pepsin

protease

(Breaks down Protein)

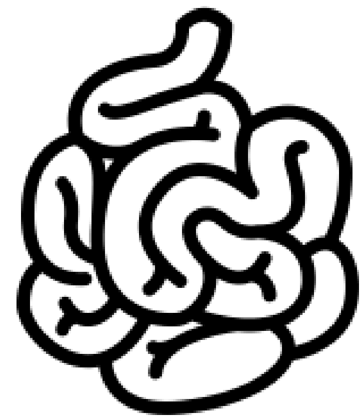


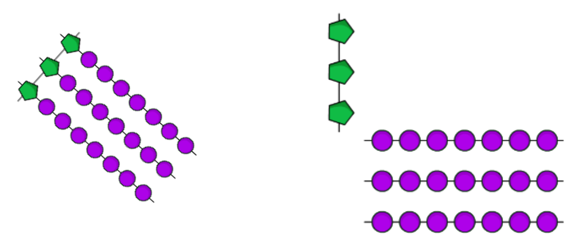


Protein → Amino acids

Lipase

(break down Lipids/fat)





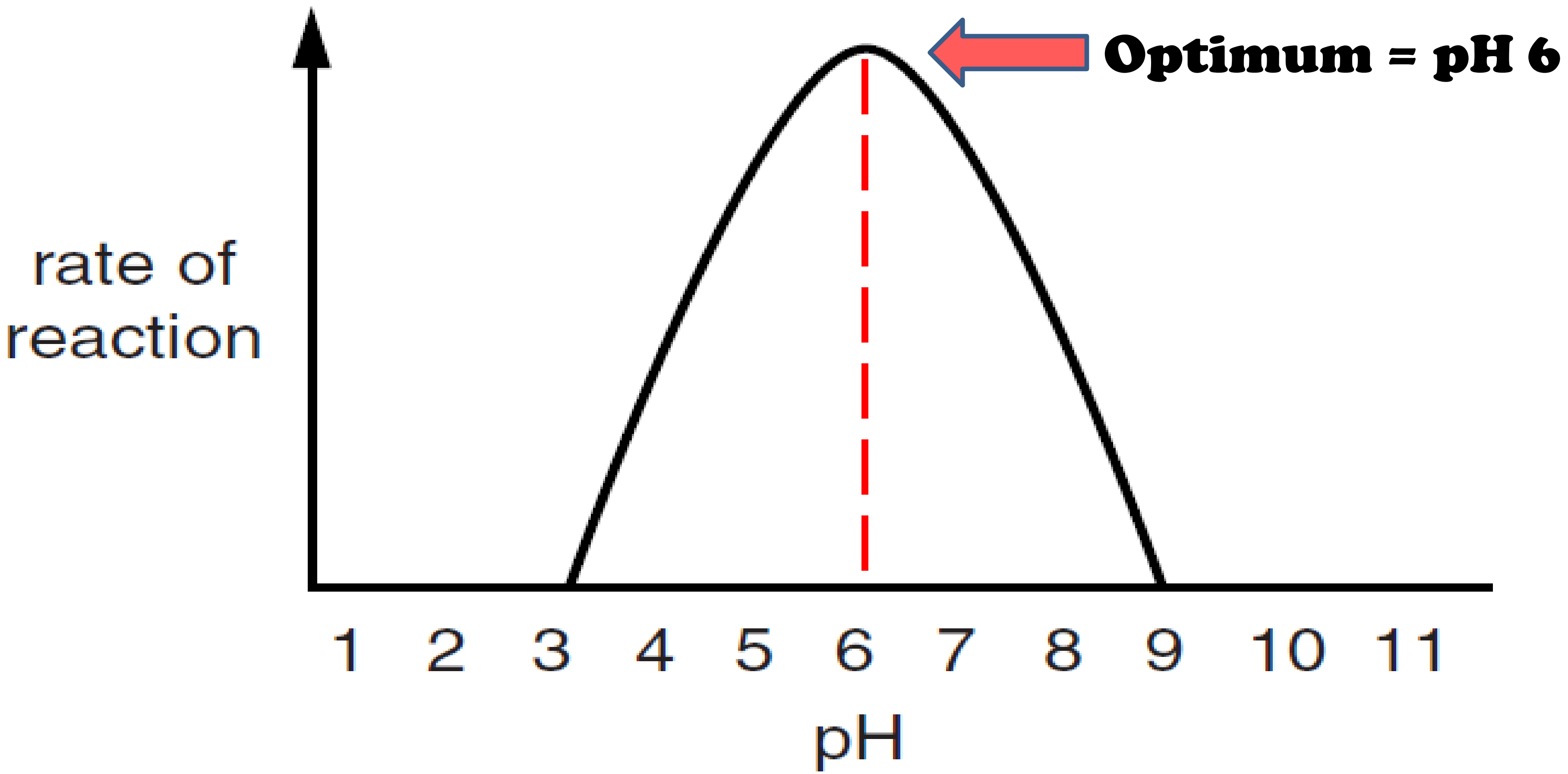
Fats → Fatty acids

All things need the correct conditions to work properly

SO DO ENZYMES!

Reactivity of Enzymes

All enzymes have optimum conditions to work in. The optimum is the condition which will produce the highest rate of reaction.

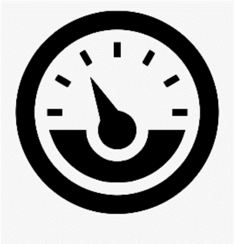


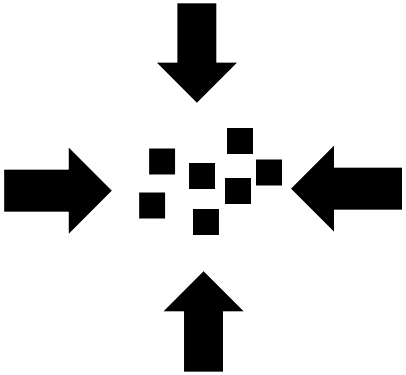
Factors affecting enzyme activity

pH

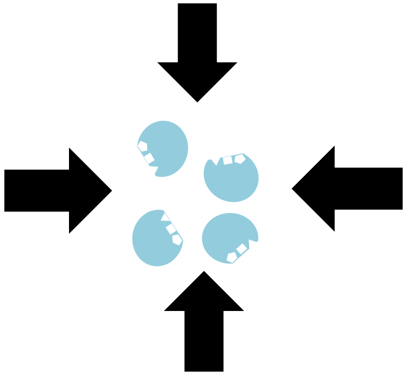
Temperature

Pressure





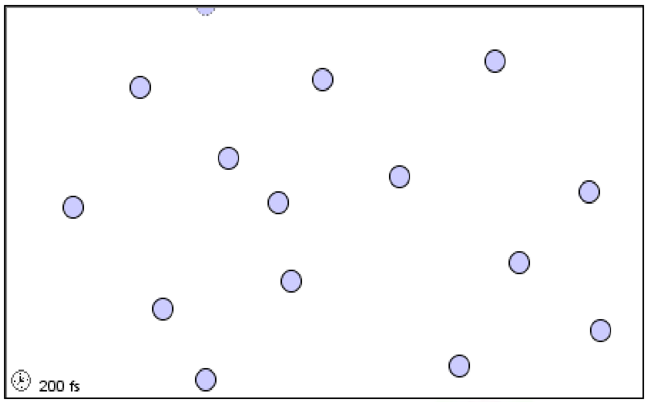
Substrate concentration

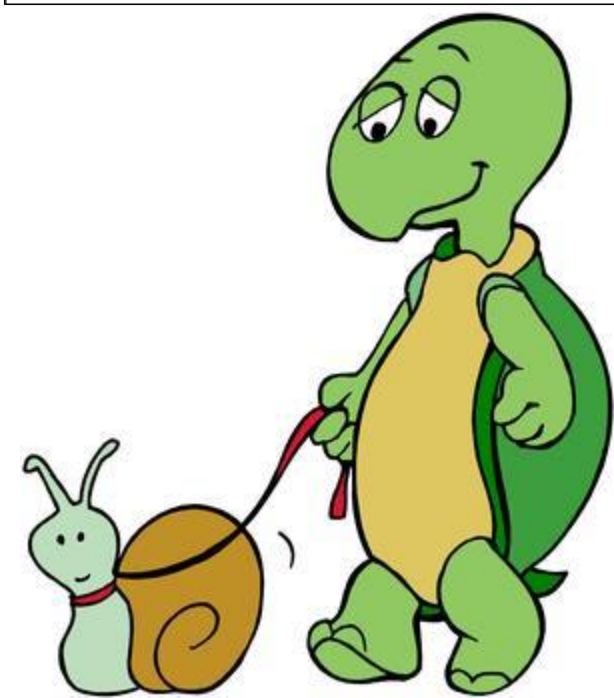


Enzyme concentration

Why is temperature important?

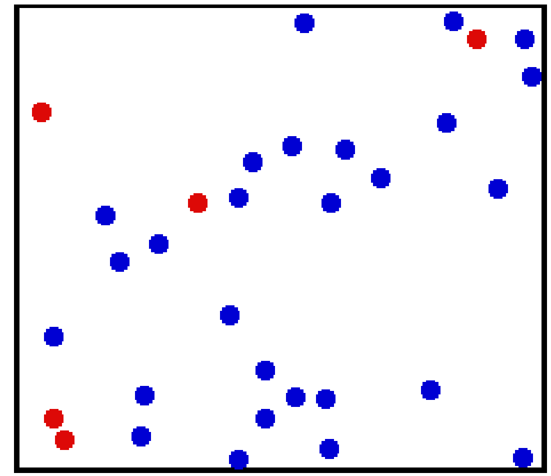
When the temperature is cold there is not enough kinetic energy (particles are moving slow).





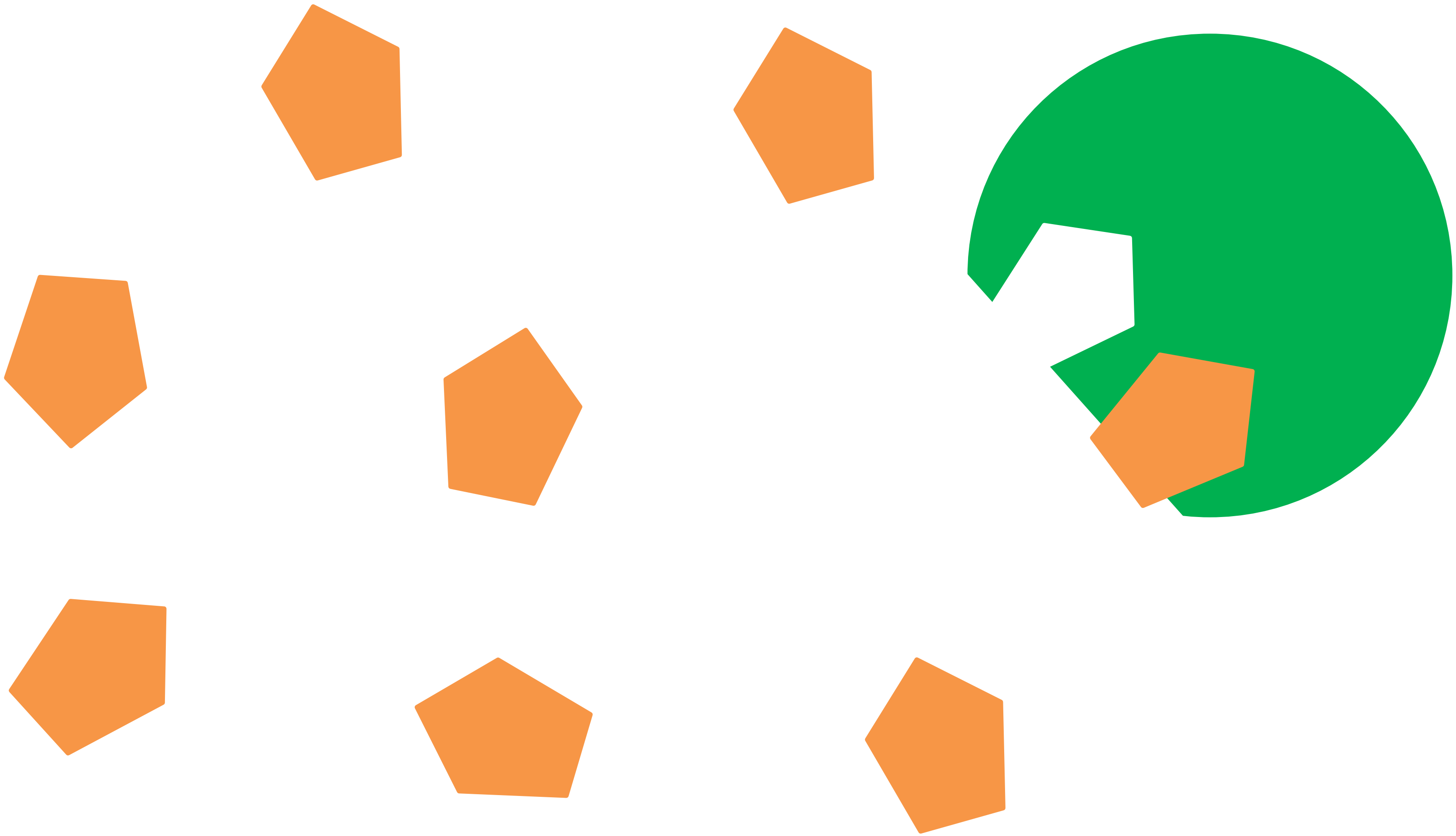
heat

Increasing temp increase the kinetic energy = more likely to react



Why is temperature important?

Fast moving particles means there is an **increased frequency of collisions** increasing the chances the **substrate joins** with the **active site.**



Denaturing

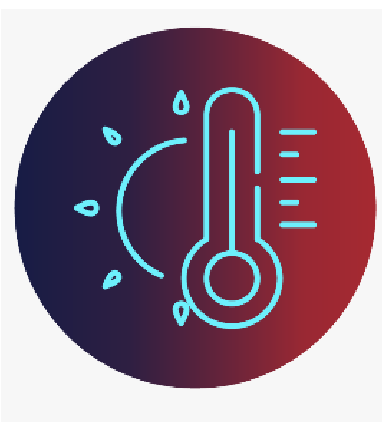
A process where ***proteins lose their shape*** *due to pH or temperature limiting or stopping an enzyme being an effective catalyst.*

normal

HIGH TEMPERATURE

Enzyme denature when too hot

denatured



The amino acid chains of the enzyme unravel so the active site changes.

Not all enzymes have an optimum of 40º. Bacteria in hot springs survive in conditions of 80º in hot springs or 0º in the deep sea. They are called **extremophiles**.

Activity: Using the data and your graph paper draw the graph to show temperature and its effect on enzyme action.

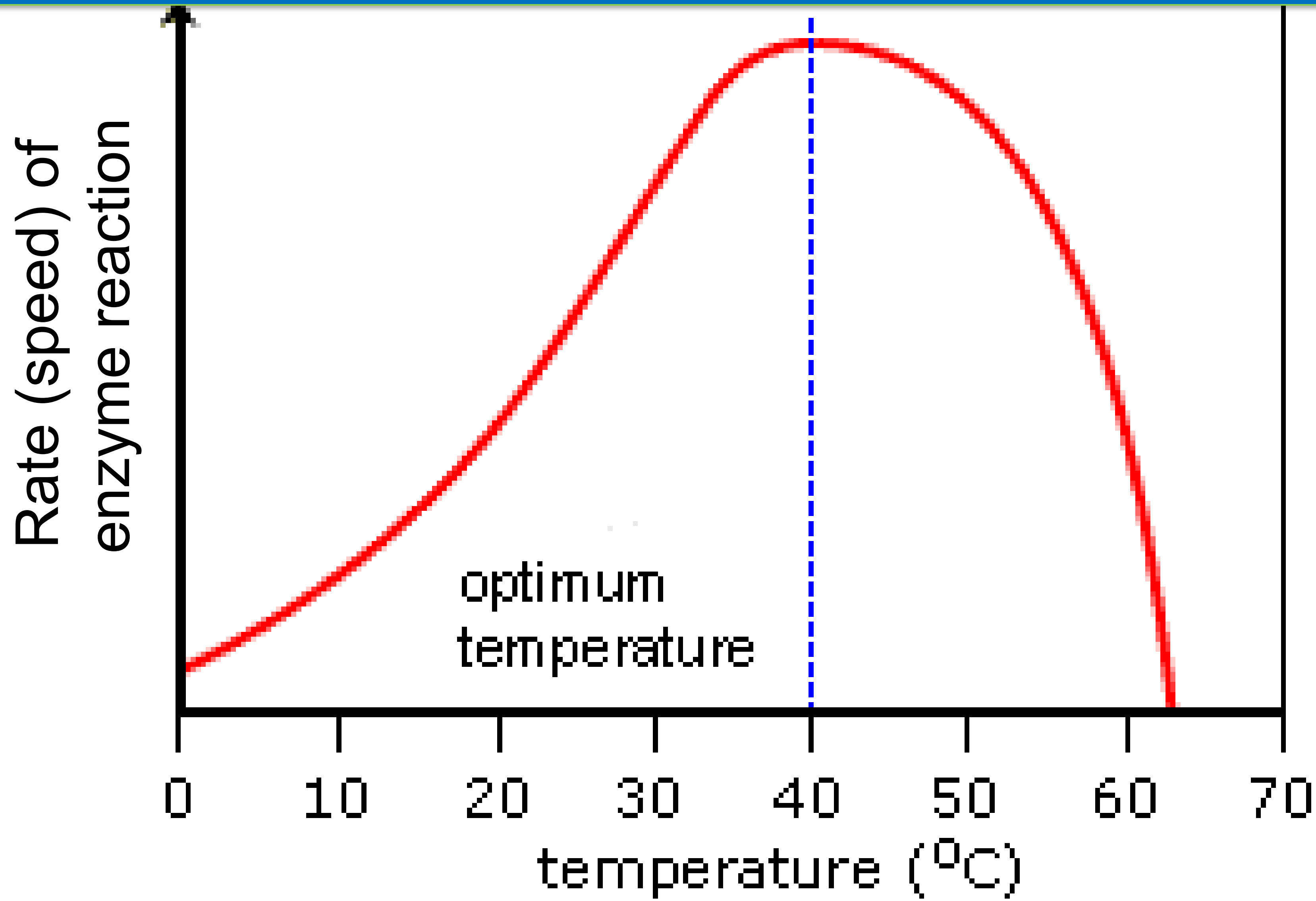
EXT: On your graph label and explain the beginning section, middle and end scientifically

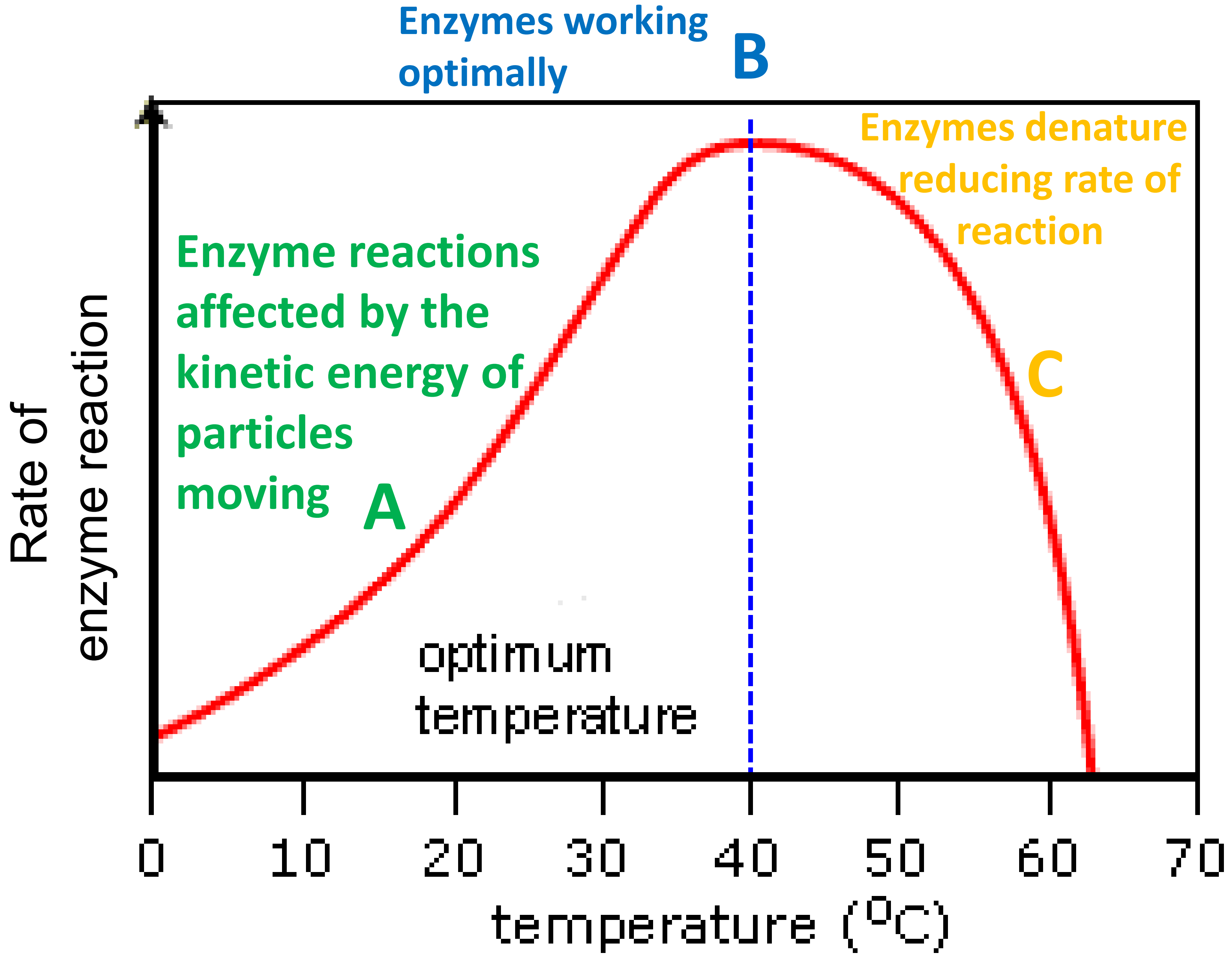
Temperature (ºC)	Rate of enzyme reaction
0	8
10	22
20	47
30	84
40	105
50	95
60	47
63	0

Key points

- Choose your scale correctly
- Label your *x* and *y axis with the variables*
- Include the units for your labels
- Plot each point accurately with and X not a ●
- A continuous line of best fit through the points. Don’t worry if not all the points fit onto the line

Review
Enzymes denature at temperatures above their optimum; normally above 40ºC.





Enzymes working optimally
B
Enzymes denature reducing rate of reaction
C
Enzyme reactions affected by the kinetic energy of particles moving
A
optimum temperature
Rate of enzyme reaction
0
10
20
30
40
50
60
70
temperature (°C)

Denaturing

*A process where **proteins lose their shape** due to pH or temperature limiting or stopping an enzyme being an effective catalyst.*

denatured

LOb: Understand the factors that affect the action of an enzyme.

Keywords: Denature, pH, temperature, optimum, reaction, extremophile

Success Criteria:

- ✓ Describe what denaturing is and how an enzyme is denatured by temperature
- ➢ Explain how pH affects enzyme action

Did you know… **Rubisco** is the most important enzyme as it allows plants to turn carbon dioxide in the air into carbon which allows plants to grow. Without this enzyme life as we know it would not exist. **Rubisco** is also the most abundant enzyme in the world.

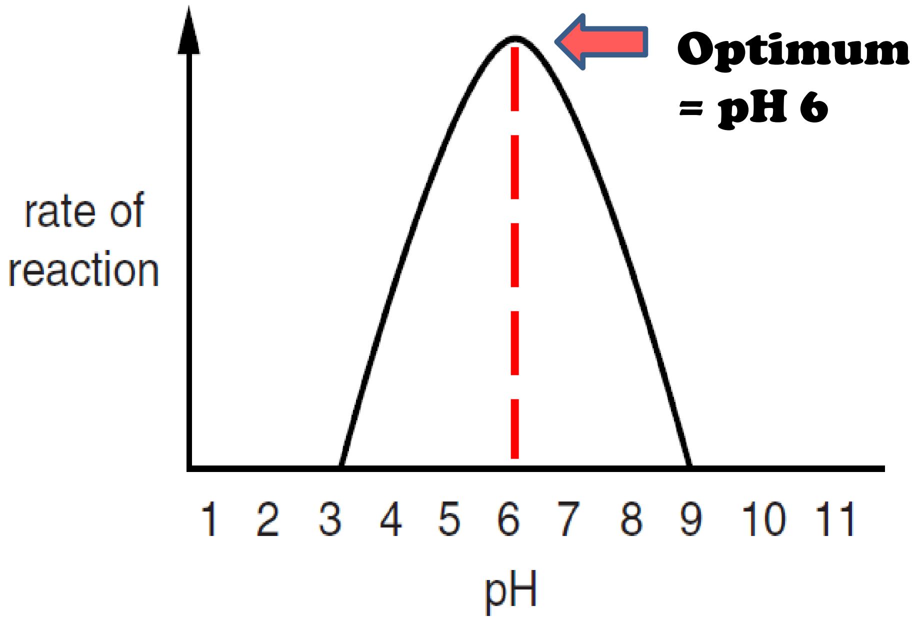
Activity: Watch this video and get ready to explain this graph

WATCH AND

LEARN

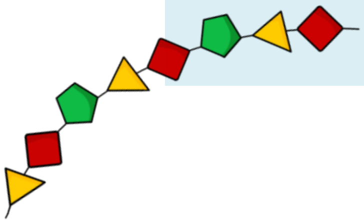
AFTERALL KNOWLEDGE IS

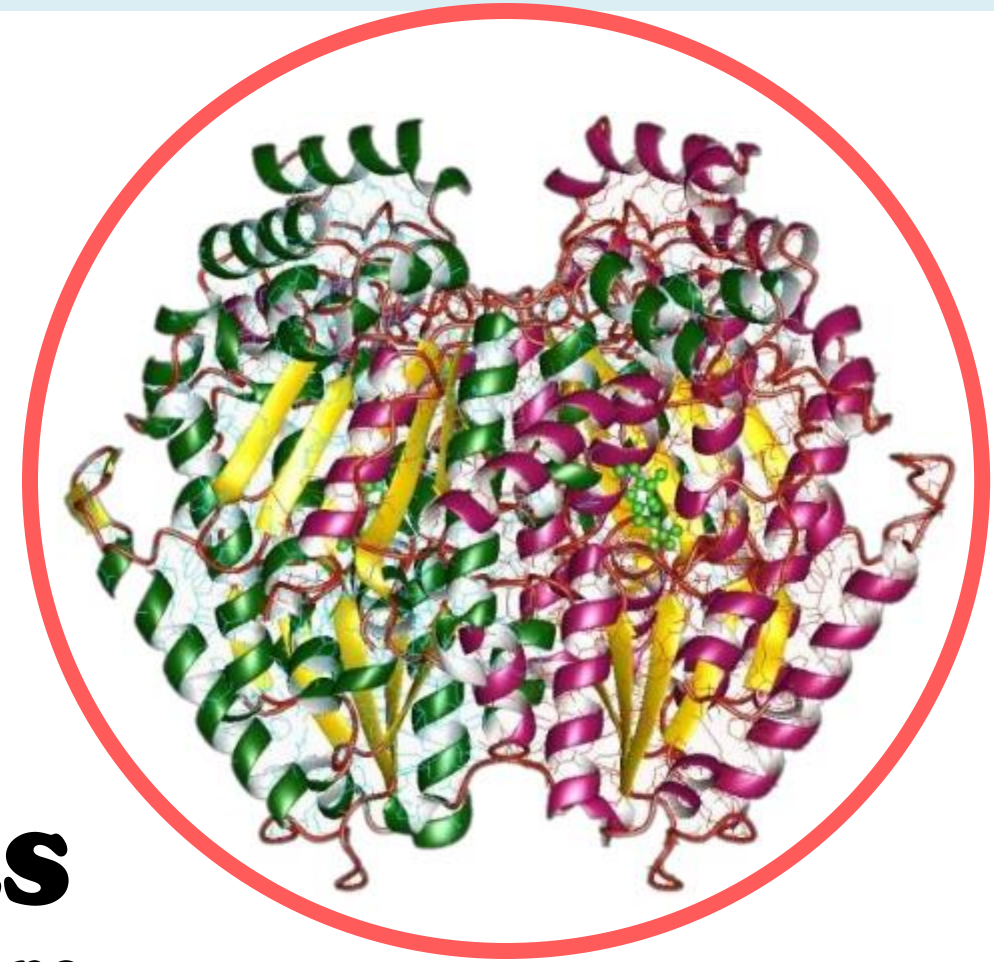
POWER!



The reason enzymes denature at some pHs is because of the **forces** that are **exerted** on their **protein structure.**

Here is a computer generated diagram of the proteins in an particular enzyme.



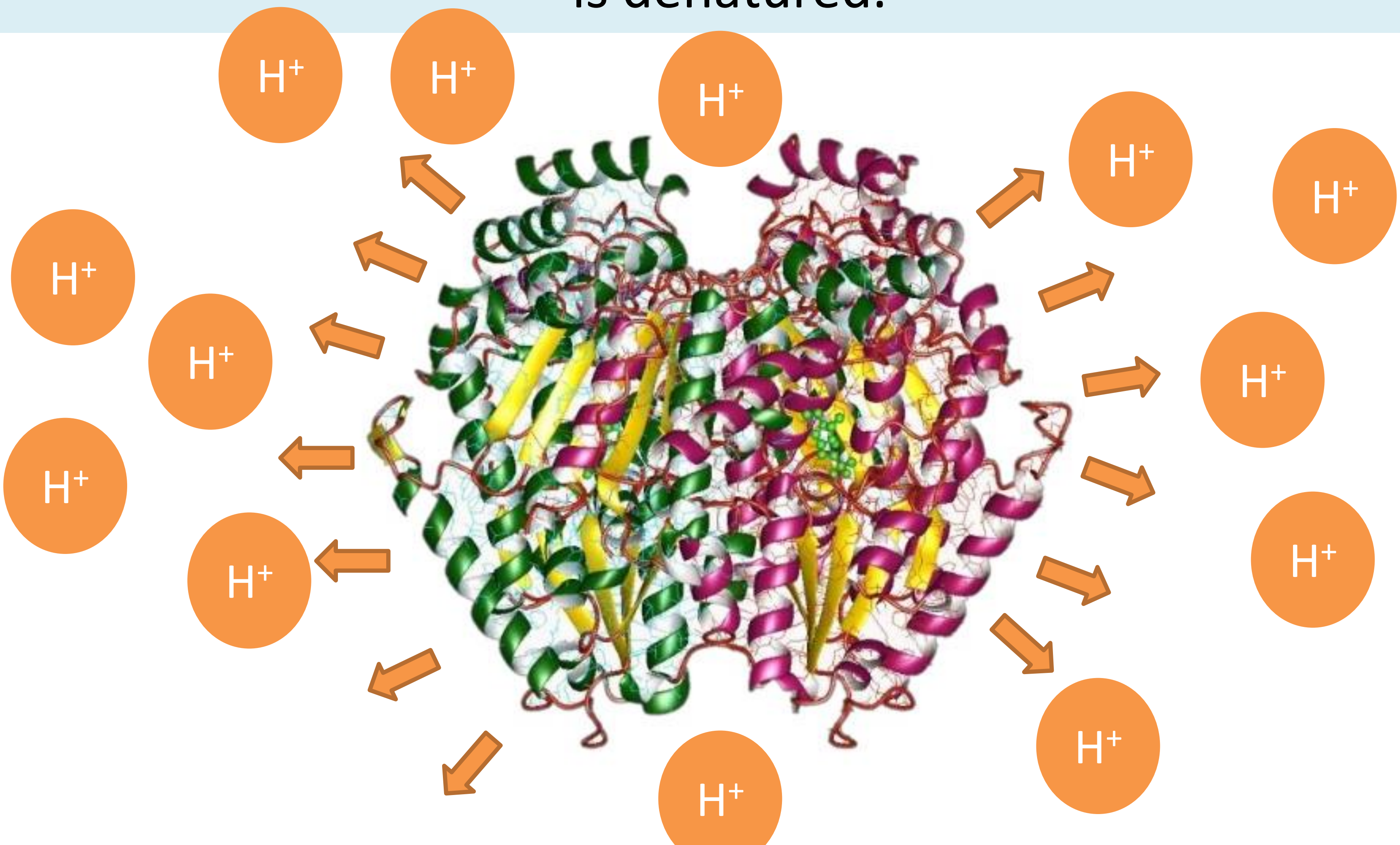


FORCES

hold the shape

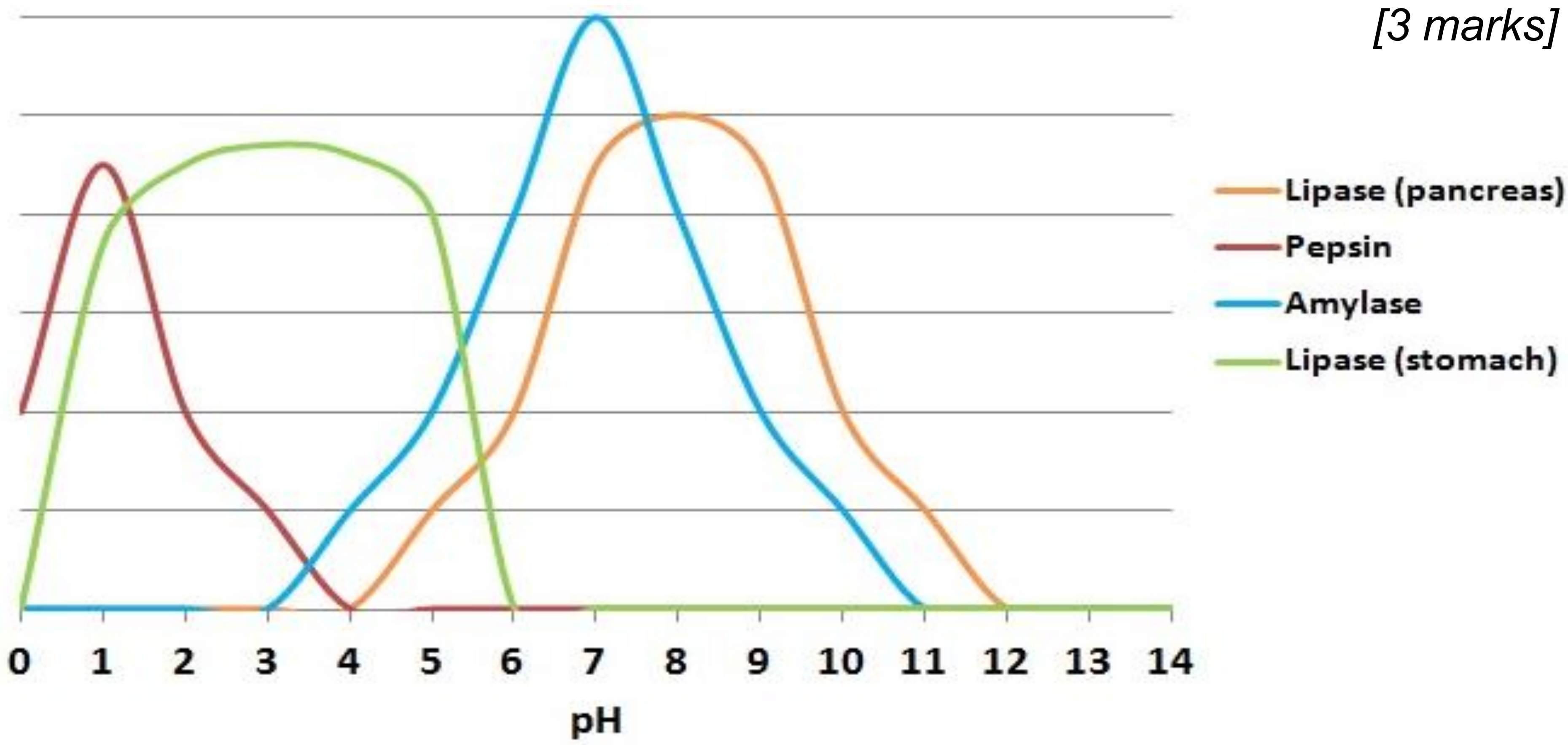
Acids have lots of H^+ ions which exert a force.

This pull on the protein structure makes the enzyme lose its form / shape. Active site can't fit the substrate. The enzyme is denatured.



Questions

1. What is the optimum pH for amylase? *[1 mark]*
2. Which enzyme has the greatest optimal range? *[1 mark]*
3. Why is pepsin's optimal pH so low? *[2 marks]*
4. Explain how pH affects enzymes differently from temperature. Make reference to the differences in the rate of activity curves on the graph. *[3 marks]*



Questions

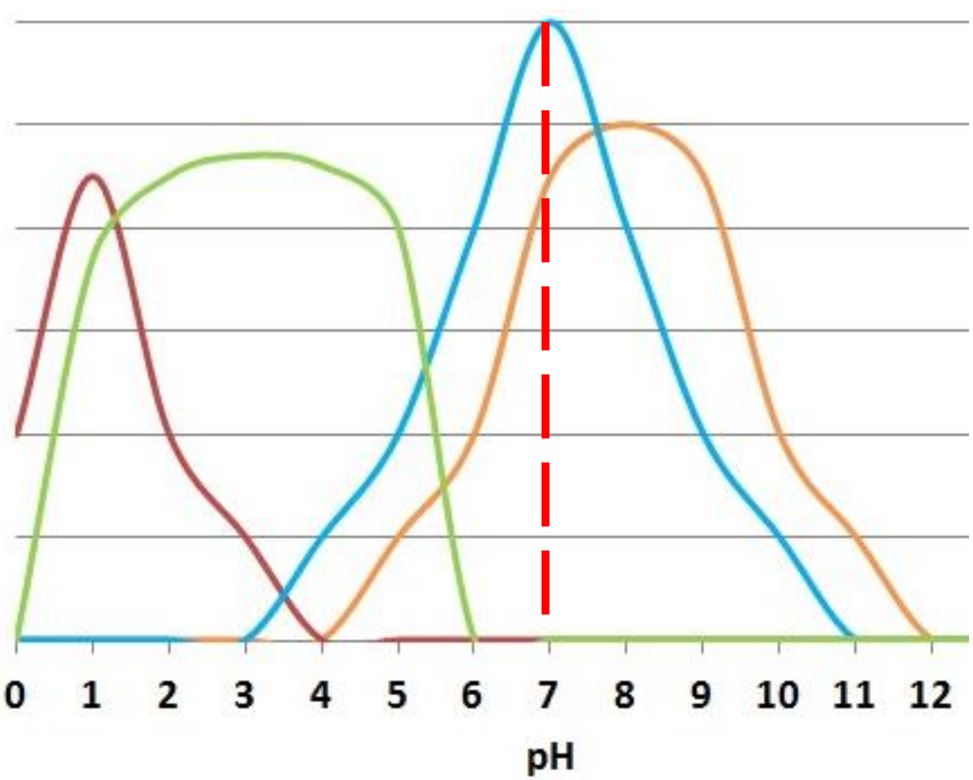
1. What is the optimum pH for amylase? **pH 7**

2. Which enzyme has the greatest optimal range? **Lipase (stomach)**

3. Why is pepsin's optimal pH so low? **Pepsin is found in the Stomach. Stomach acid lowers the pH and pepsin works well.**

4. Explain how pH affects enzymes differently from temperature. Make reference to the differences in the rate of activity curves on the graph.

[3 marks]



Enzymes do not denature at low temperature only at high [1]

pH affects enzymes by exerting or failing to exert forces on their structure causing them to change shape and denature [1]

pH curves have steep lines on both sides of optimal [1]

LOb: Understand the factors that affect the action of an enzyme.

Keywords: Denature, pH, temperature, optimum, reaction, extremophile

Success Criteria:

- ✓ Describe what denaturing is and how an enzyme is denatured by temperature
- ✓ Explain how pH affects enzyme action

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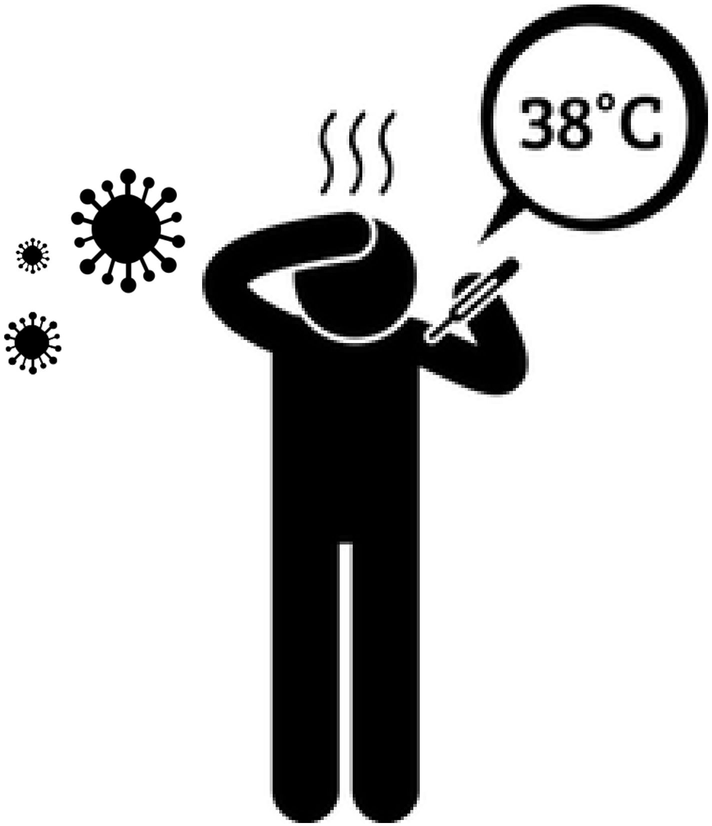
Application discussion

When a have an infectious disease you may **'get a temperature'.** This is a protection mechanism used by your body as many **microorganism can't reproduce at higher temperatures.** However, people always try and **bring down the temperature** of an ill person. Explain why and discuss is it the best thing to do.



keyfacts

- **Factors** that effect enzymes: **pH, temperature, substrate and enzyme concentration, pressure, surface area**

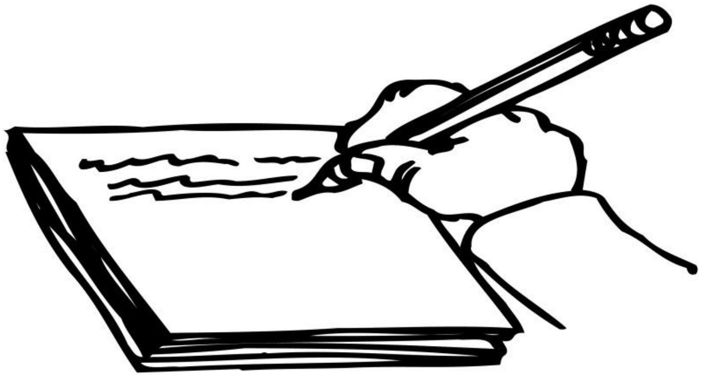
- Each enzyme has an **optimal condition.**

- **Temperature** effects enzyme activity. **Cold temp = low kinetic energy reduced frequency of collisions.** .

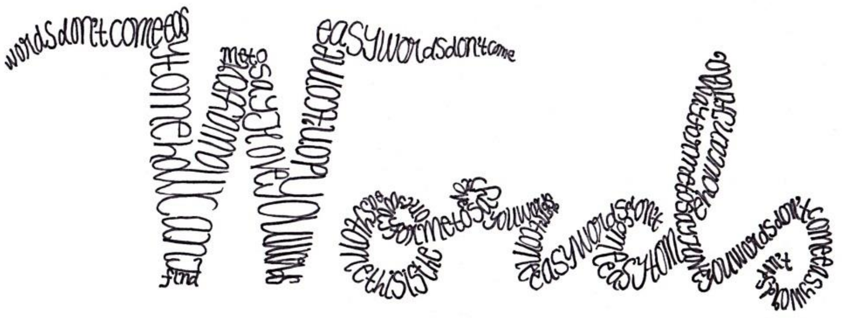
- **Denaturing** at **high temperature** or a **pH range** means that the **structure of proteins** has been **irreparably damaged** which **stops the enzyme** from being able to **catalyse** a reaction.

- **H^+ ions exert forces** on enzyme protein structures which are either too strong or to weak to what they are used to so they denature.

Plenary: Simplify



Firstly write two sentences to summarise your learning



Then reduce to 5 words

Then reduce to 1 keyword

Question Number		Indicative Content	Mark
QWC	***6(d)**	A description to include some of the following points Temperature • (temperature) too low – not enough energy to make reactions occur (fast enough) • reference to optimum temperature • optimum for most (humans) - 37°C • over 37°C changes enzyme shape / changes active site shape of enzyme • therefore rate of reaction decreases / stops • enzymes denatured (if temperature too high) pH • optimum pH – around 7.3 / 6 to 8 for most enzymes • specific optimum quoted eg pepsin – pH 2 to 3 • pH either side of optimum – changes the shape of the enzyme / shape of the active site • therefore rate of reaction decreases / stops • enzymes denatured (if pH too high / too low) substrate / enzyme concentration • higher concentrations faster reactions • due to more collisions • until maximum rate reached / all enzymes being used	**(6)**
Level	**0**	No rewardable content	
1	**1 – 2**	• a limited description of how temperature OR pH OR substrate concentration affects the rate of enzyme action • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 – 4**	• a simple description of two or more factors OR a detailed description of one factor • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 – 6**	• a detailed description of at least two factors • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	